

SE_LAB_2

Name:-Prithvish Adhikari

SRN:-PES1UG24AM206

Table of Epics and User Stories

Type	Key	Epic / User Story	Parent Epic	Priority	Points
Epic	GTESEL2-1	Expense Recording and Splitting	-	Highest	-
Epic	GTESEL2-2	Trip and Participant Management	-	Highest	-
Epic	GTESEL2-3	Multi-Currency Conversion	-	High	-
Epic	GTESEL2-4	Expense Summary and Settlement Tracking	-	High	-
Epic	GTESEL2-5	Debt Settlement Optimization	-	Highest	-
Story	GTESEL2-6	Create a group trip	GTESEL2-2	Highest	3
Story	GTESEL2-7	Convert foreign-currency expenses	GTESEL2-3	High	5
Story	GTESEL2-8	Add and manage travelers	GTESEL2-2	High	2
Story	GTESEL2-9	Record a group expense	GTESEL2-1	Highest	3
Story	GTESEL2-10	Retrieve the daily exchange rate	GTESEL2-3	High	3
Story	GTESEL2-11	Split an expense unevenly	GTESEL2-1	Highest	5
Story	GTESEL2-12	Calculate each traveler's net balance	GTESEL2-5	Highest	5
Story	GTESEL2-13	Generate the minimum-transaction settlement plan	GTESEL2-5	Highest	8
Story	GTESEL2-14	View the trip expense summary	GTESEL2-4	Medium	3
Story	GTESEL2-15	Record a completed settlement payment	GTESEL2-4	High	3

Summary : 5 epics, 10 genuine user stories, 40 story points; all graded items are Done.

Epic and User Story Explanations

Epic 1: Expense Recording and Splitting (GTESEL2-1)

Enable group travelers to record shared expenses and divide costs using equal or uneven shares.

Story 1.1: Record a Group Expense (GTESEL2-9)

As a Group Traveler, I want to enter an expense with its payer, amount, date, participants, and currency, so that it is included in the trip account.

Points: 3

Story 1.2: Split an Expense Unevenly (GTESEL2-11)

As a Group Traveler, I want to assign unequal shares to participants, so that the split represents what each person actually owes.

Points: 5

Epic 2: Trip and Participant Management (GTESEL2-2)

Allow a Trip Admin to create a trip, select its base currency, and manage all participating travelers.

Story 2.1: Create a Group Trip (GTESEL2-6)

As a Trip Admin, I want to create a trip and choose its base currency, so that group expenses can be managed consistently.

Points: 3

Story 2.2: Add and Manage Travelers (GTESEL2-8)

As a Trip Admin, I want to add or remove travelers, so that the correct participants are included in the trip.

Points: 2

Epic 3: Multi-Currency Conversion (GTESEL2-3)

Convert foreign-currency expenses into the trip's base currency using applicable daily exchange rates.

Story 3.1: Convert Foreign-Currency Expenses (GTESEL2-7)

As a Group Traveler, I want foreign expenses converted into the trip's base currency, so that all balances are comparable.

Points: 5

Story 3.2: Retrieve the Daily Exchange Rate (GTESEL2-10)

As a Trip Admin, I want the applicable daily exchange rate to be retrieved and stored, so that currency conversions remain accurate and auditable.

Points: 3

Epic 4: Expense Summary and Settlement Tracking (GTESEL2-4)

Provide a clear trip-expense summary and track settlement payments completed between travelers.

Story 4.1: View the Trip Expense Summary (GTESEL2-14)

As a Group Traveler, I want to view a summary of expenses, contributions, and balances, so that I can verify the trip account.

Points: 3

Story 4.2: Record a Completed Settlement Payment (GTESEL2-15)

As a Group Traveler, I want to mark a settlement payment as completed, so that outstanding balances and settlement status are updated.

Points: 3

Epic 5: Debt Settlement Optimization (GTESEL2-5)

Calculate each traveler's net balance and generate a settlement plan using the fewest practical peer-to-peer payments.

Story 5.1: Calculate Each Traveler's Net Balance (GTESEL2-12)

As a Group Traveler, I want to see how much each traveler owes or should receive, so that the group's financial position is clear.

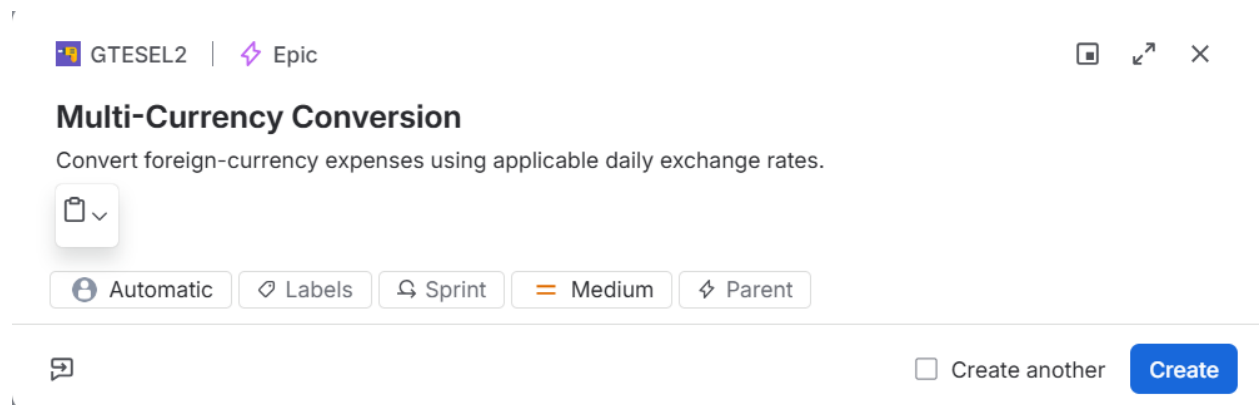
Points: 5

Story 5.2: Generate the Minimum-Transaction Settlement Plan (GTESEL2-13)

As a Group Traveler, I want an optimized settlement plan, so that the group can settle all debts using the fewest practical payments.

Points: 8

Screenshots of Epics and User Stories



GTESEL2

Epic

Expense Recording and Splitting

Allow travelers to record expenses and divide them using equal or uneven shares.

Automatic

Labels

Sprint

Medium

Parent

Create another

Create

Epic

>

Expense Recording and Splitting

>

Trip and Participant Management

>

Multi-Currency Conversion

>

Expense Summary and Settlement Tracking

>

Debt Settlement Optimization

Screenshots of Epics

GTESEL2

Story

Convert foreign expenses

As a Group Traveler, I want foreign expenses converted into the trip's base currency so that all balances are comparable.

Automatic

Parent

Sprint

High

Labels

Create another

Create

GTESEL2

Story

Record an expense

As a Group Traveler, I want to enter an expense with its payer, amount and currency so that it is included in the trip account.

Automatic

Parent

Sprint

Highest

Labels

Create another

Create

Screenshots of some of the user stories being created

☐	GTESEL2 Sprint 3	Add dates	(5 of 6 work items visible)	16	0	0	Start sprint	...
	...							
☒	GTESEL2-7	Convert foreign-currency expenses		Multi-Currency Co...	To Do	5		
☒	GTESEL2-9	Record a group expense		Expense Recordin...	To Do	3		
☒	GTESEL2-10	Retrieve the daily exchange rate		Multi-Currency Co...	To Do	3		
☒	GTESEL2-6	Create a group trip		Trip and Participan...	To Do	3		
☒	GTESEL2-8	Add and manage travelers		Trip and Participan...	To Do	2		

Unordered grouping in first Sprint

☐	GTESEL2 Sprint 3	Add dates	(5 of 6 work items visible)	16	0	0	Start sprint	...
	...							
☒	GTESEL2-6	Create a group trip		Trip and Participan...	To Do	3		
☒	GTESEL2-9	Record a group expense		Expense Recordin...	To Do	3		
☒	GTESEL2-7	Convert foreign-currency expenses		Multi-Currency Co...	To Do	5		
☒	GTESEL2-10	Retrieve the daily exchange rate		Multi-Currency Co...	To Do	3		
☒	GTESEL2-8	Add and manage travelers		Trip and Participan...	To Do	2		
Create								

Ordered the grouping based on priority in first Sprint

Start Sprint



5 work items will be included in this sprint.

Required fields are marked with an asterisk *

Sprint name *

GTESEL2 Sprint 3

Duration *

1 week

Start date *

9/1/2026 2:48 PM

Date format: MM/DD/YYYY. Time format: e.g. 1:00 PM.

End date *

9/8/2026 2:48 PM

Sprint goal

Cancel

Start

Starting the first sprint

GTESEL2 board

SummaryTimelineActive sprintsBacklogCalendarListFormsDevelopmentDocs

Search boardVersionEpicLabelQuick filtersComplete sprintGroup: Story

TO DO 1

Record a group expense

Expense Recording and Splitting

GTESEL2-9

Create

IN PROGRESS 1

Create a group trip

Trip and Participant Management

GTESEL2-6

DONE 3

Convert foreign-currency expenses

Multi-Currency Conversion

GTESEL2-7

Add and manage travelers

Trip and Participant Management

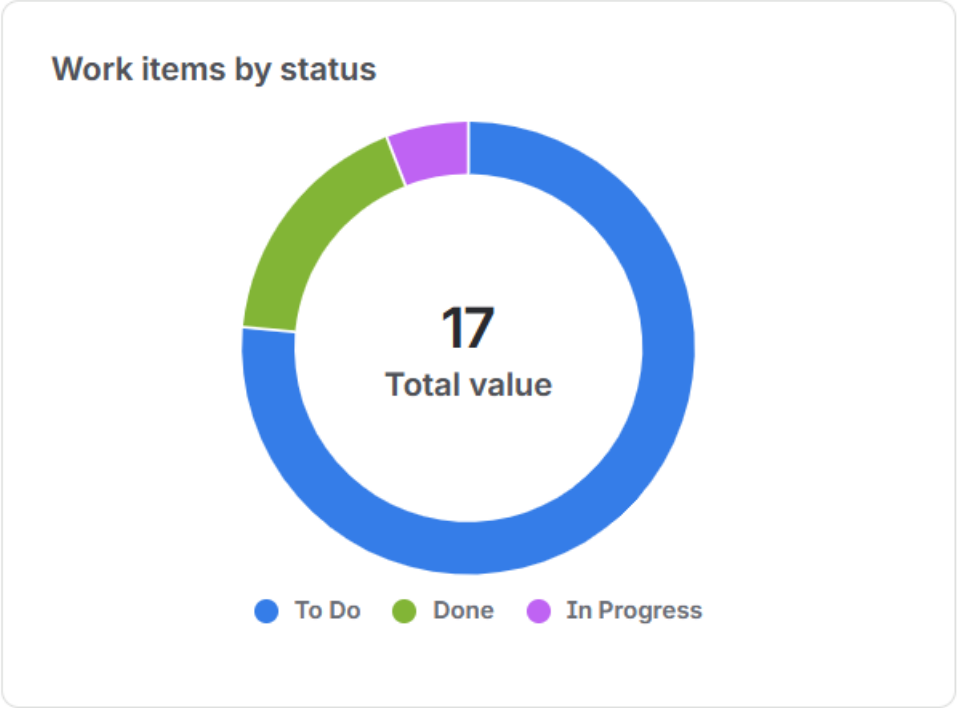
GTESEL2-8

Retrieve the daily exchange rate

Multi-Currency Conversion

GTESEL2-10

Completing the first sprint



Pie Chart for Work Item’s Status

☐ GTESEL2 Sprint 4 📅 Add dates (7 work items)

27

3

0

☒ GTESEL2-6 Create a group trip	Trip and Participan... In Progress ▾	3
☒ GTESEL2-12 Calculate each traveler's net balance	Debt Settlement O... To Do ▾	5
☒ GTESEL2-11 Split an expense unevenly	Expense Recordin... To Do ▾	5
☒ GTESEL2-14 View the trip expense summary	Expense Summary... To Do ▾	3
☒ GTESEL2-15 Record a completed settlement payment	Expense Summary... To Do ▾	3
☒ GTESEL2-13 Generate the minimum-transaction settlement plan	Debt Settlement O... To Do ▾	8
☒ GTESEL2-9 Record a group expense	Expense Recordin... To Do ▾	3

Unordered grouping in second sprint

☐	GTESEL2 Sprint 4  Add dates (7 work items)	27	3	0	Sta
☒	GTESEL2-6 Create a group trip	Trip and Participan...	In Progress ▾	3	
☒	GTESEL2-12 Calculate each traveler's net balance	Debt Settlement O...	To Do ▾	5	
☒	GTESEL2-11 Split an expense unevenly	Expense Recordin...	To Do ▾	5	
☒	GTESEL2-13 Generate the minimum-transaction settlement plan	Debt Settlement O...	To Do ▾	8	
☒	GTESEL2-9 Record a group expense	Expense Recordin...	To Do ▾	3	
☐	☒ GTESEL2-15 Record a completed settlement payment 	Expense Summary...	To Do ▾	3	
☒	GTESEL2-14 View the trip expense summary	Expense Summary...	To Do ▾	3	

Ordered the grouping based on priority in second sprint

Start Sprint



7 work items will be included in this sprint.

Required fields are marked with an asterisk *

Sprint name *

GTESEL2 Sprint 4

Duration *

1 week

Start date *

9/1/20269:09 PM

Date format: MM/DD/YYYY. Time format: e.g. 1:00 PM.

End date *

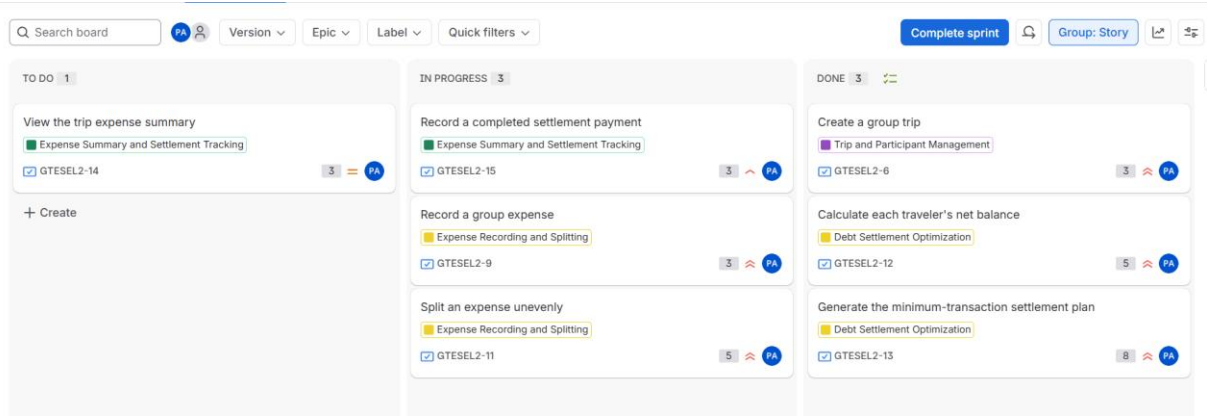
9/8/20269:09 PM

Sprint goal

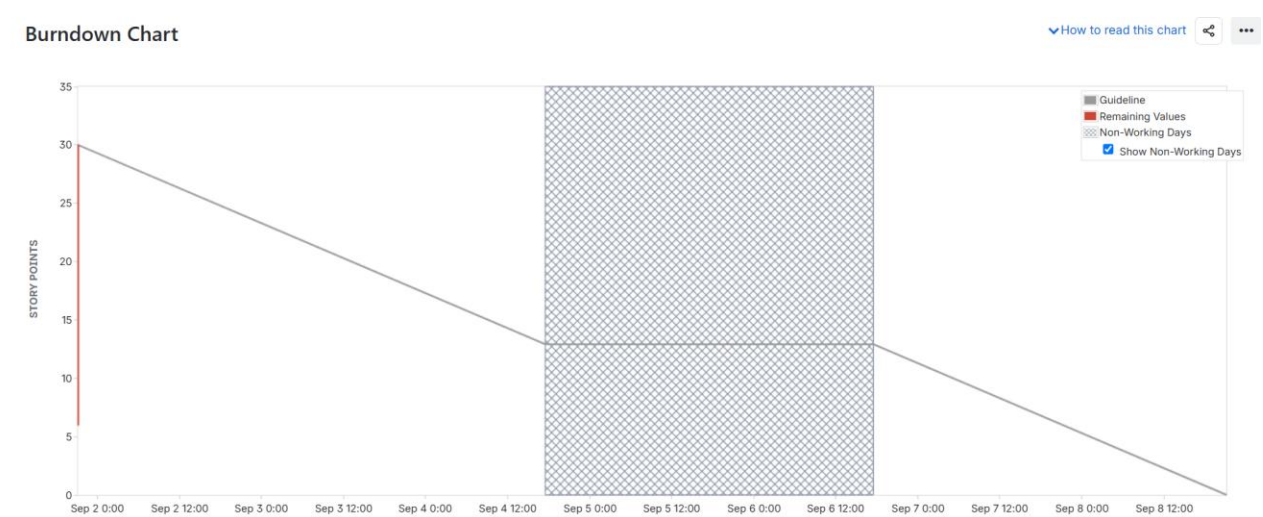
Cancel

Start

Starting second sprint



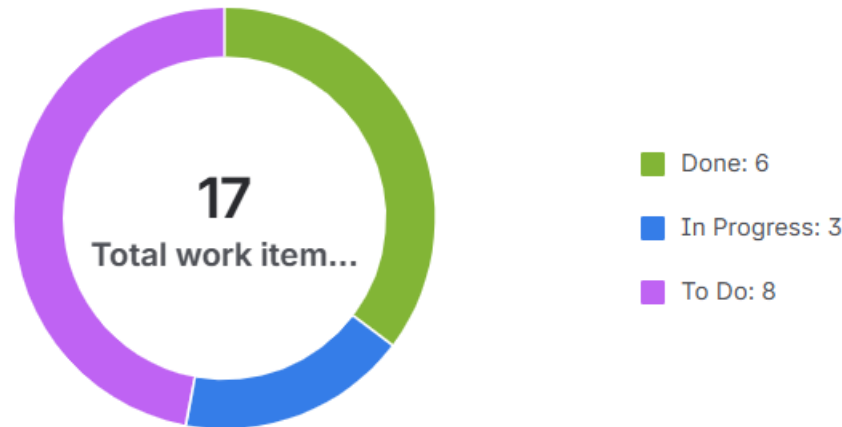
Completing the second sprint



Burndown chart after completing two sprints

Status overview

Get a snapshot of the status of your work items. [View all work items](#)



Pie chart of work status after two sprints



Prithvish Adhikari

15 hours ago

Epic completed: both linked stories—group-expense recording and uneven splitting—are Done and validated against their acceptance criteria.



Prithvish Adhikari

15 hours ago

Epic completed: trip creation and traveler-management stories are Done, covering base-currency setup, administration, participant addition/removal, and duplicate prevention.



Adding several comments in activity of epics

Planning Poker Analysis

Planning Poker was used to estimate the relative effort, complexity, uncertainty, and testing requirements of the user stories in the Group Travel Expense Simplification Engine. The Fibonacci scale of 2, 3, 5, and 8 story points was used. Each story was discussed before selecting an estimate, and the final value represented the team's consensus rather than the amount of time required to complete the work.

The “Add and Manage Travelers” story was assigned 2 points because it mainly involves straightforward participant-management operations with limited business logic. Stories such as “Create a Group Trip,” “Record a Group Expense,” “Retrieve the Daily Exchange Rate,” “View the Trip Expense Summary,” and “Record a Completed Settlement Payment” were assigned 3 points because they require moderate form handling, validation, data storage, and integration with other system components.

The “Convert Foreign-Currency Expenses,” “Split an Expense Unevenly,” and “Calculate Each Traveler's Net Balance” stories were assigned 5 points. These stories involve additional calculation rules, currency accuracy, rounding, balance validation, and multiple edge cases. The “Generate the Minimum-Transaction Settlement Plan” story received the highest estimate of 8 points because it requires an optimization algorithm, handling multiple debtor-

creditor combinations, and extensive correctness testing to ensure all balances are settled using the fewest practical transactions.

Group Reflection

This activity helped us understand how epics, user stories, priorities, story points, and sprints are managed in Jira. Planning Poker helped us estimate stories based on complexity and effort. We organized ten stories into two balanced sprints of 21 and 19 points and tracked them from To Do to Done. We learned that clear requirements, proper estimation, teamwork, and regular progress comments are important for successful Agile development.

